Table II Correction: Cause, Evidence, and Replacement Results

1. Cause and evidence

Item	Earlier implementation / available alternatives	Current implementation and evidence
Graph input	The data directory contains multiple graph exports with different sizes and embedding dimensions.	The current `main.py` loads `miRNA_disease_network.json`: 90,631 records, 90,206 unique directed pairs, 8,977 nodes, and 256-dimensional embeddings.
Duplicate edges	The older function does not first simplify duplicate graph edges.	The current function applies `dgl.to_simple`. The selected JSON contains 425 duplicate records beyond its 90,206 unique pairs.
Negative sampling	The older implementation samples non-edges among all nodes and does not seed sampling; negatives are not restricted to miRNA–disease candidates.	The current function samples unique miRNA × disease non-edges with seeded, disjoint fold partitions.
Fold and validation protocol	The older implementation uses shuffled folds without a fixed random state and sets validation to 80% of the fold's training indices.	The current protocol seeds folds and uses 10% validation by default, selects epochs using validation AUC, and selects the classification threshold using validation F1.
Message passing	In the older path, validation/test scoring uses the full graph, allowing held-out edges to inform node embeddings.	The current path removes held-out validation/test edges from the message-passing graph; test scoring uses train plus validation edges.
Run settings	The historical command, code revision, and preprocessing manifest for the published Table II values are not retained with the table values.	The checked-in result tag records attention dropout 0.1, four heads, learning rate 0.01, two layers, predictor input size 2, output dimension 256, and 200 epochs.

The loader builds node IDs and feature rows in NetworkX node iteration order, then attaches those rows to the DGL graph. The selected JSON has embeddings on every node and all are 256-dimensional, so the zero-vector fallback is not invoked for this input.

2. Changes in the revised protocol

1. Held-out edges and message passing. The earlier path could compute validation/test node embeddings using the full graph, including evaluated edges. The current path removes validation/test edges, including reverse copies, from the training message-passing graph. Test embeddings use train plus validation edges. This can lower ranking metrics by removing information leakage.

2. Negative examples. The earlier code sampled non-edges among all nodes, although the task is miRNA–disease prediction. The current code samples unique miRNA × disease non-edges and keeps negative partitions separate across training, validation, and test.

3. Training edges and model selection. The earlier function assigns 80% of training indices to validation, leaving about 16% of all positive edges for fitting in a five-fold split. The current protocol uses 10% validation within the train-plus-validation partition, restores the epoch with best validation AUC, and selects the classification threshold on validation F1.

4. Duplicates and randomness. The current path simplifies duplicate edges before partitioning and seeds folds, model initialization, and negative sampling. Seeding improves repeatability but cannot recover an undocumented historical seed.

5. Metric definitions. The revised path uses probabilities for threshold metrics and selects the threshold on validation; average precision is calculated consistently. Per-fold bootstrap half-widths are reported separately from the standard deviation across folds. These choices can affect F1, precision, recall, mAP, and uncertainty.

3. Corrected five-fold results

One complete five-fold run is shown below. All folds use balanced test sets with a 1:1 positive-to-negative ratio of miRNA–disease pairs. Each fold's uncertainty is the half-width of a 95% stratified bootstrap interval over test pairs. The final row is the mean and sample standard deviation across the five folds. F1, accuracy, precision, and recall use the threshold selected by validation F1.

Fold	AUC (%)	F1 (%)	Accuracy (%)	Precision (%)	Recall (%)	mAP (%)
1	93.70 ± 0.26	88.68 ± 0.31	88.22 ± 0.34	85.33 ± 0.43	92.31 ± 0.39	92.20 ± 0.42
2	95.44 ± 0.26	92.02 ± 0.28	92.08 ± 0.28	92.73 ± 0.34	91.31 ± 0.42	95.83 ± 0.34
3	95.22 ± 0.24	90.91 ± 0.29	90.83 ± 0.30	90.09 ± 0.41	91.75 ± 0.40	95.29 ± 0.33
4	82.57 ± 0.46	81.53 ± 0.34	79.60 ± 0.40	74.49 ± 0.44	90.03 ± 0.44	78.18 ± 0.63
5	95.07 ± 0.25	91.31 ± 0.29	91.44 ± 0.29	92.73 ± 0.35	89.94 ± 0.42	95.40 ± 0.34
Mean ± SD	92.40 ± 5.54	88.89 ± 4.30	88.43 ± 5.15	87.07 ± 7.66	91.07 ± 1.05	91.38 ± 7.52

The stored fold log gives best epochs 89, 132, 79, 9, and 97. The metrics in the table agree with the per-fold values in the CSV and log. Fold 4 is substantially weaker than the others; the across-fold SD therefore reflects meaningful fold variation and should not be presented as a small standard error.